Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80% Retake your highest score.

Next Item →

Optimizing your AI requests

Overview

In this activity, you will use Google AI Studio to improve the responses you get from a chatbot. You will learn how to change the model's thinking level, create a reusable system prompt, and test your changes. Thinking will be the key to control the AI's output, just as make sure your applications are helpful, accurate, and efficient.

AI optimization

Scenario

Imagine you work for a company that creates online learning materials. They want to use AI to help create new content, like quizzes and lesson summaries. However, the current AI responses are inconsistent. Sometimes they are too simple, and sometimes the model is too slow. You need to experiment to find the right model, settings, and prompts to get high-quality results every time.

Steps

Step 1: Choose your task and write a starting prompt

First, decide which task you want to work on. This will help you focus your experiments.

- Go to Google AI Studio and, as **google**, **Sign** in.
- On the home menu, click **Chat With Models**.
- Select the model **Gemini 3 Pro Preview** on the right-hand side.
- Choose **one** of the following use cases and write a simple prompt to fit the task and box.
  - **Use Case 1: Create an email template**
    - Prompt: Write **Imagine you are a** a trainer expert, **teaching** and **to a** group **parttime** relevant **topic** and **Explain** the **basic** concepts.
  - **Use case 2: Help brainstorm ideas**
    - Prompt: Write **Write** a short, **exciting** account **media** post about the **launch** of a new **electric** bicycle.
  - **Use Case 3: Generate quiz questions**
    - Prompt: Write **Create** ten **multiple-choice** quiz **questions** based on this **Text:** . **The** correct **answer** should **be** A. **Format** the **text** so **that** each of the other **answers** in our output **options** appear **on** separate **lines.**
- On the right side of the screen, you will see the settings. For now, leave the model as **Gemini 3 Pro Preview** and all other settings at their default values.
- Select the **Run** button and look at the output. This is your "baseline" result—the result you will try to improve.

Step 2: Experiment with "Thinking" level

The setting on the right called **Thinking level** can dramatically change the AI's response. Let's see it now.

Conceptual note: Thinking level controls **how much reasoning the model does before answering**.

- **Low thinking:** Faster, more direct responses, best for simple tasks and straightforward instructions.
- **High thinking:** More deliberate reasoning, best for complex tasks that require planning, multi-step logic, or careful analysis.

Find the Thinking level setting.

- **Run 1 (Low Thinking):** Set Thinking level to **Low**. Click **Run** again. Notice how the output is likely faster and more concise.
- **Run 2 (High Thinking):** Set Thinking level to **High**. Click **Run**. Notice how the output may be more thorough, better structured, and more careful with multi-step reasoning.

Tip: Use **Low** for quick, simple, summary, and recipe transformations. Use **High** for complex reasoning, planning, and tricky analysis tasks.

Adjusting the AI's "Thinking" level

1. **You're testing the same prompt twice in AI Studio and want the second run to produce a more thorough, multi-step answer instead of a fast one. Which Thinking level should you choose?** Select the best answer. 1.1 point
   - High Thinking
   - Low Thinking
   - Either one - Thinking level only changes the writing style
   - Other Thinking

   ✅ Nice work.
   Correct. High Thinking is better for deeper reasoning and multi-step logic, even if it's slower.

2. **You're creating a bond rate chatbot for a line of customers you evaluate the bots to deeper reasoning to handle edge cases like unexpected phrasing. Which Thinking level should you choose to evaluate the quality of bond rate responses?** 1.1 point
   - Thinking level — High
   - Thinking level — Medium
   - Thinking level — Low
   - Thinking level — Off / Disabled

   ✅ Nice work.
   Correct. High Thinking level generally encourages the model to spend more effort reasoning and exploring alternatives, which can lead to more nuanced and surprising ideas (often with higher latency/cost).

Step 3: Compare different models

Now, let's compare how the with Gemini Flash for speed and cross.

1. Set the **Thinking level to High.**
2. Make sure your original prompt is still in the box.
3. Under the **Model** dropdown menu on the right, select **Gemini 3 Pro Preview**. Click **Run** and save the output.
4. Set the Thinking level to low and run the prompt again.
5. Now, change the model to **Gemini 3 Flash Preview** and **Thinking Level** to **High**. Click **Run** again.
6. Set the Thinking level to low and run the prompt again.

Look at the two outputs. Gemini Flash is much faster and cheaper, making it good for simpler or repeated tasks.

4. **Your company wants to quickly analyze thousands of short customer feedback comments on any day. The main priority is an speed and low cost, while the complexity of the task is relatively low.**
   Which model combination most closely matches this? Select the best answer. 1.1 point
   - Both models would work equally well
   - The Gemini Pro model
   - Neither model is suitable for this task
   - Pro is better than Flash is not

   ✅ Nice work.
   Correct. Gemini Flash is designed for speed and cost efficiency. It's the ideal choice for high-frequency, low-complexity tasks like summarizing short comments.

Step 4: Iterate on your results

Good developers keep notes on their experiments. This helps them remember what worked best and allows them to compare results objectively.

- At **Step 3** fill in the table below to create your own copy of this **AI Distillation Re-Skinned.sg**.
- Copy **[Click here to make your own copy in Google Docs.]** 🚀
- **Instructions:** The link will take you to **"Make a copy"** Once accepted, the document will be saved to your personal Google Drive file you modify.

What you MUST do in the worksheet

Fill in the table you created in the document with your observations from the steps above.

Common MISTAKE: Not writing down your results. It is easy to forget which setting produced which output. Taking good notes is key skill.

Step 5: Reflect and Guide on the best approach

Look at your notes in the table. Based on your experiments, answer these questions for yourself.

1. For your chosen task, which model worked better, Pro or Flash? Why?
2. What was the best Thinking Level setting for your task?
3. How could you make your original prompt even better to get an even more useful response?

Advanced challenge (Optional)

Based on your reflection, try to improve your original prompt. Add more context, constraints, or a specific persona (e.g., "Write the email as if you were the voice of a enthusiastic co-pilot"). Run the improved prompt with your chosen model and parameter settings. Does the output improve?

4. **You are building a chatbot that must provide consistent and reliable answers to frequently asked questions. When you test an answer you notice the chatbot's responses vary in how detailed they are—sometimes short and brief, other times more step-by-step. What is the most likely cause of this difference?** Select the best answer. 1.1 point
   - The prompt is too short
   - The Thinking level is different than it, Flight.
   - The API key is incorrect.
   - The model is unsupported entirely.

   ✅ Nice work.
   Correct. The Thinking level controls how much reasoning the model does—Low tends to be faster and more direct, while High tends to be more thorough and multi-step.

5. **You want your AI to always act as a professional travel agent and always use a friendly, helpful tone, no matter what new prompts a user types in. Where is the best place to put these "permanent" instructions in AI Studio?** Select the best answer. 1.1 point
   - System Instructions
   - The Model dropdown menu
   - In the Advanced Challenge section
   - At the end of every user prompt

   ✅ Nice work.
   Correct. System Instructions (box above you're define persona, tone, or set of rules that apply to the entire interaction without having to repeat them in every prompt.

Success checklist

- You have tested at least two different Thinking Level settings.
- You have compared the output from both of the selected Flash.
- You have filled out a documentation table with your results.

Key points

- **Consistency is controllable by Thinking Level:** Setting **Thinking Level to Low** helps produce more consistent, direct responses for factual tasks like classifications, while **High** may introduce reasoning and variability in how the answer is explained.
- **Trade-offs:** The **Flash model** is best when speed and low cost are your main goals, the **Pro model** is best when reasoning and high quality are required.

Example solution

Here's a quick example demonstrating the optimization process for the social media post use case.

- **Selected use case:** Create a social media post.
- **Baseline prompt:** Write a short, exciting social media post about the launch of a new electric bicycle.

Test number 1 (baseline)

Model used: Gemini Pro

Thinking level: High

Prompt used: Prompt above

Output quality (example result):

Say goodbye to traffic and hello to freedom! 🚲💨

Introducing the **[Insert Bike Name]**—where sleek design meets serious power. Whether you're tackling your morning commute or enjoying a weekend adventure, it's effortlessly by your side. ✨

The electric revolution is here, and you're ready to ride!

👉 **Get yours at [Link]**

#RideLife #FreeElectric #FutureOfTravel #NewLaunch

Test number 2

Model used: Gemini Pro

Thinking level: Low

Prompt used: Prompt above

Output quality (example result):

Ready to leave traffic in the dust?

Meet the [Bike Name]—your new favorite way to fly through the city. 🚲 Instant torque, all-day battery, and style for miles.

Your commute just became the best part of your day. 🙌

👉 Link in bio to learn more now!

#EBikeLife #NewLaunch #UrbanMobility #RideLife #ElectricBikes

Test number 3

Model used: Gemini Flash

Thinking level: High

Prompt used: Prompt above

Output quality (example result):

Ready to upgrade your commute? Here is an exciting, high-energy social media post for the launch of a new electric bicycle.

Headline: The Future of Two Wheels is HERE! 🚲💨

Stop sweating against the wind and start gliding through your city. We are officially LIVE! [Insert Bike Name] is the ultimate, sleekest, and most powerful e-bike we've ever built. 🚀

Why you'll love it:

🔋 **Infinite Range:** Our new high-density battery lets you go further on a single charge than ever before.

⚡ **AI-Powered Drift:** Intelligent sensors that adjust to your gear—so even a sudden pace changes.

🪶 **Ultra-Lightweight:** At the weight of a pro-grade motor in a throw weight, you'll forget it's electric.

📱 **Smart App Tech:** Turn-by-turn navigation and anti-theft tracking right from your phone.

Whether you're tackling your morning commute or exploring hidden trails, the [Model Name] makes every ride feel like a victory lap. 🏁

Grab yours now and get [Insert Early Bird Discount, e.g., 15% OFF] for a limited time!

👉 [Link to Website/Shop]

#EBike #LaunchWeek #ElectricRevolution #SmartCycling #GreenMobility #CommuterBike #NewArrival #ConquerTheMile #RideBeyondTheLimit

Test number 4

Model used: Gemini Flash

Thinking level: Low

Prompt used: Prompt above

Output quality (example result):

1. **THE WAIT IS OVER. THE FUTURE IS ELECTRIC.** ⚡

Say hello to the all-new [Model Name]—your ultimate ride has arrived! 🚲💨

Whether you're tackling your morning commute or hitting the weekend trails, our [Traits, Softer, and with zero emissions. We've combined cutting-edge tech with sleek, lightweight design to give you the ride, for other.

✨ **Key Features:**

🔋 **Massive Range:** Go up to [X] miles on a single charge.

🚀 **Lightning Speed:** Hold the power of our [X] motor.

📱 **Smart Integration:** Track your ride and lock your bike straight from your app.

Ready to ride? Be the way you know! The first 50 orders get a spectacularly fast discount. 🎁

👉 **Shop the Revolution at the link in our bio!**

#RideTheWave #EcoWarrior #[ModelName] #UrbanMobility #GreenCommute #FutureOfTransport #WeekendVibes

Reflection: We've created a task like this, **High Thinking** can produce a far more thoughtful output, while **Low Thinking** is faster and more direct. **Gemini 3 Pro** handles a task like this **High Thinking** can be extremely precise and still creepy enough for more drafts. If I needed to generate hundreds of emails, I'd use **Gemini Flash with Low Thinking** for speed and cost-efficiency, then switch to **High Thinking** only for the final "hero" versions.

Next steps

Great job! You now know how to test and improve your AI requests and models. In the next module, you will learn how to take these ideas from AI Studio and turn them into code for your applications.